

Vectors

Key terms

vector, magnitude, scalar, linear combination, dot product

KSBs: K13, S11, B1, S53

8.1 Vectors

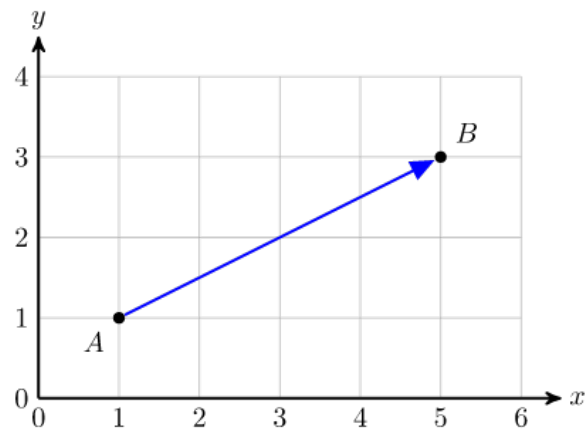
Vectors are a way of expressing quantities that cannot be described by a set of data, or a single point in a data set with

This could involve a . It can also be interpreted geometrically.

Vectors are sometimes thought of as being an 'arrow' in n -dimensional space. We can describe the location of a point in space using (or in 3D). A vector describes the as an arrow from one to the other.

For example, consider a vector in 2D space: there is a distance in one direction, and a distance in another, so a cannot carry this information.

Vectors were originally used in geometry, but have since become important in data science, computer graphics, engineering and physics.



Definition 8.1 - Vector

A **vector** is a collection of numbers, called written as an which defines a

We usually denote vectors by subscript numbers, written as a row or column:

like $\mathbf{v}$ or $\mathbf{a}$, and the entries in the vectors with

To make it clear what we're talking about, the character representing a vector is written in boldface when typeset, e.g., $\mathbf{a}$ or underlined when written by hand, e.g., $\underline{a}$.

Exercise 8.1. In the statement below, underline x whenever it refers to the vector.

“The vector x is defined as $x = (x_1, x_2, x_3)$. The x -coordinate of x is x_1 .”

We use $\mathbf{0}$ to denote the **zero vector** from a point a to itself.

This can be thought of as the vector $a - a$, or the vector

8.1.1 Vector Magnitude

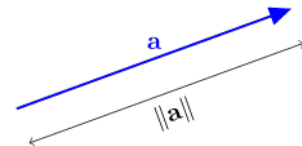
The **magnitude** of a vector $\mathbf{a}$ is the dimensions, this can be calculated using an extended version of it.

between the head and tail of $\mathbf{a}$. In two and in higher dimensions we can use

Definition 8.2 - Vector Magnitude)

In 2D, the **magnitude** of a vector $\mathbf{a} = (a_1, a_2)$, denoted by $\|\mathbf{a}\|$, is calculated as

For a general vector $\mathbf{a} = (a_1, a_2, \dots, a_n)$, $\|\mathbf{a}\|$ is given by



For any two vectors $\mathbf{a}, \mathbf{b}$ and any scalar k , the following properties are satisfied:

- $\|\mathbf{a}\| \geq 0$
- $\|\mathbf{a}\| = 0 \iff \mathbf{a} = \mathbf{0}$

Example 8.1. Let $\mathbf{v} = (2, 7, 1)$. Then the magnitude of $\mathbf{v}$ can be calculated as:

Exercise 8.2. Calculate the magnitudes of the following vectors:

- (a) $\mathbf{p} = (3, 4)$
- (b) $\mathbf{u} = (1, 2, 3)$
- (c) $\mathbf{v} = (5, -12, 0)$
- (d) $\mathbf{w} = (1, 0, 1)$

8.1.2 Arithmetic operations on vectors

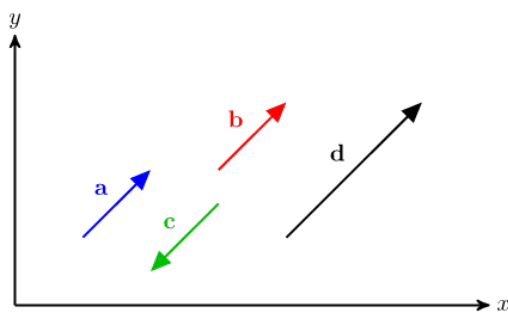
Definition 8.3 - Vector Equality

Two vectors are said to be equal if $\mathbf{a} = (a_1, a_2, \dots, a_n)$ and $\mathbf{b} = (b_1, b_2, \dots, b_n)$ respectively, are equal if and only if the elements which correspond co-ordinate-wise are equal, i.e.,

In other words,

Informally, this happens if and only if two vectors

and



Here, the vectors $\mathbf{a}$ and $\mathbf{b}$ point in the same direction and have the same magnitude, so we can say that

The vector $\mathbf{c}$ has

but points in

so

The vector $\mathbf{d}$ points in the same direction as $\mathbf{a}$ but has a larger magnitude - so

Definition 8.4 - Addition of two vectors

The sum of two vectors $\mathbf{a}$ and $\mathbf{b}$ can be calculated as

A geometric representation of the sum $\mathbf{a} + \mathbf{b}$ is shown in the figure.

We place the vector $\mathbf{a}$, and then position the vector $\mathbf{b}$ so

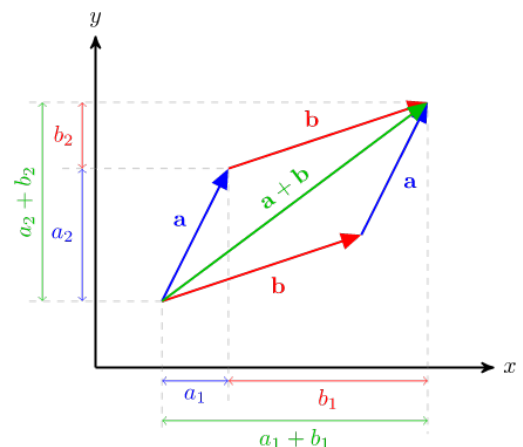
The endpoint of $\mathbf{b}$ is

Note that the order in which we add the two vectors

- if we placed $\mathbf{b}$ first, then $\mathbf{a}$, we would get to the same point. This is known as the

For two vectors $\mathbf{a}$ and $\mathbf{b}$,

This property is called the



Example 8.2. Let $\mathbf{a} = (1, 0, 2)$ and $\mathbf{b} = (2, -3, 1)$. Then

Exercise 8.3. Let $\mathbf{u} = (2, 3)$, $\mathbf{v} = (3, -2)$ and $\mathbf{w} = (1, 6)$. Write down

- (a) $\mathbf{u} + \mathbf{v}$
- (b) $\mathbf{u} + \mathbf{w}$
- (c) $\mathbf{v} - \mathbf{w}$

8.1.3 Scalar multiplication

Definition 8.5 - Scalar

A scalar is a single numerical value, which can be a

We can multiply a vector by a scalar to get

Definition 8.6 - Scalar Multiplication

The product of a vector $\mathbf{a}$ by a scalar k is given by

Multiplying a vector by a scalar has the effect of
with a magnitude of

to produce a vector

The product of a vector $\mathbf{a}$ and a scalar k has the following properties:

- $k\mathbf{a}$ is a vector parallel to $\mathbf{a}$, i.e.
- $0\mathbf{a} =$
- $1\mathbf{a} =$
- $\|k\mathbf{a}\| =$
- If $k > 0$ then $\mathbf{a}$ and $k\mathbf{a}$

whereas if $k < 0$ then $\mathbf{a}$ and $k\mathbf{a}$



Example 8.3. Let $\mathbf{v} = (1, 2, 3, 4)$ and $k = 6$. Then

Exercise 8.4. Calculate $k\mathbf{v}$, where $\mathbf{v} = (2, 3, 4)$ and

- (a) $k = 2$
- (b) $k = 1.5$
- (c) $k = 1$
- (d) $k = 0$

8.1.4 Linear combination of vectors

Definition 8.7 - Linear combination of vectors

Let $\mathbf{v}, \mathbf{u}_1, \dots, \mathbf{u}_m$ be n -dimensional vectors such that:

for scalars $\alpha_1, \alpha_2, \dots, \alpha_m$. Such a sum is called a

Note

α is the lowercase Greek character *alpha* and is equivalent to 'a' in the Latin alphabet.

For example:

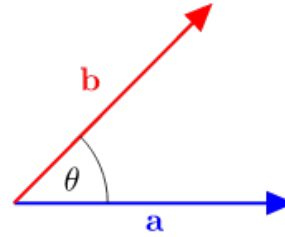
Here we have expressed $(2, 0, 7)$ as a linear combination of the vectors $(1, 5, -1)$ and $(0, -10, 9)$. Later we will see how this can help us to solve systems of related linear equations.

8.1.5 Dot Product

Definition 8.8 - The dot product

The **dot product** (also known as the **scalar product**) of two vectors $\mathbf{a}$ and $\mathbf{b}$ is defined as

The **dot product** of two vectors $\mathbf{a}$ and $\mathbf{b}$ can be defined in geometric terms as



where θ is

Roughly, we can think of the dot product $\mathbf{a} \cdot \mathbf{b}$ as a scalar that describes

If two vectors point product to be larger than the dot product of two vectors pointing in

we would expect their dot

From the definition of the dot product, we can say:

- The dot product produces a
- The dot product of two perpendicular vectors is zero, and we write
- The dot product of two parallel vectors $\mathbf{a}$ and $\mathbf{b}$ is
- $\mathbf{a} \cdot \mathbf{b} =$
- $\mathbf{a} \cdot \mathbf{a} =$, and if $\mathbf{a} \cdot \mathbf{a} = 0$ then .

Example 8.4. Let $\mathbf{a} = (1, 2, 3)$ and $\mathbf{b} = (3, -1, 0)$. The dot product of $\mathbf{a}$ and $\mathbf{b}$ is:

Exercise 8.5. Which pairs of the following vectors are perpendicular?

$$\mathbf{u} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \quad \mathbf{v} = \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix} \quad \mathbf{w} = \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}$$

Function: SUMPRODUCT (E/S)

Recall that the `=SUMPRODUCT()` function will take as inputs two ranges of the same size, and will return the sum of the pairwise products. This is the same calculation as the dot product.

8.2 Vectors in Data Science

The dot product is a useful tool to combine two vectors, and has several uses in data science contexts.

Example 8.5. Recall that data can be presented in a

To find the mean of the data in the table, we can

and add them together to get the value of $\sum x$, then

If we stored these two sets of data as vectors, we would have

$$\mathbf{v} = (1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7 \ 8 \ 9) \quad \mathbf{f} = (1 \ 0 \ 1 \ 3 \ 2 \ 0 \ 1 \ 1 \ 2)$$

Then

Value	Frequency
1	1
2	0
3	1
4	3
5	2
6	0
7	1
8	1
9	2

This can be divided by the sum of the frequencies to get the mean of the dataset,

8.2.1 Covariance

Recall that the covariance of two random variables X and Y is given by

Theorem 8.1 - Covariance and dot product

Let X and Y be two random variables, both with a mean of zero. Then

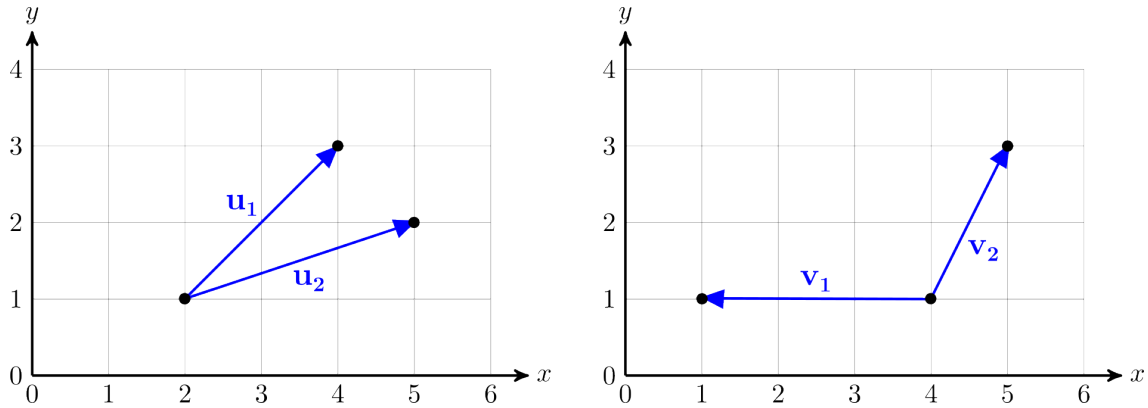
That is, the $\mathbf{v} \cdot \mathbf{f}$ (considered as vectors), divided by

Indeed, if the means of both datasets are zero, we have

This is just the $\frac{\mathbf{v} \cdot \mathbf{f}}{\sum f_i}$ on top of the fraction.

8.2.2 Vector Agreement

Recall that the dot product of two vectors can be thought of as a measure of how much the two vectors



In the image, u_1 and u_2 are pointing in while v_1 and v_2 are pointing in
Recall that if two vectors are perpendicular (at right angles), their dot product
 is

We would expect $u_1 \cdot u_2$ to be and $v_1 \cdot v_2$ to be

If we have data points in n dimensions representing n variables of linked data, if two vectors representing data points are they are likely to have and this can allow us to categorise how
 similar two observations are.

This can be applied in any number of dimensions, and is used in machine learning, image recognition and many other applications.